Optimizing IT Support Team Performance Using Analytics

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Course: Virtual Internship 7.0(Data Visualization)

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Abstract:

This project analyzes IT support ticket data to evaluate performance, identify inefficiencies, and improve resolution processes using data analytics and visualization tools.

Introduction:

IT support teams play a crucial role in maintaining system efficiency. This project focuses on analyzing ticket data to improve response time and service quality.

Objectives:

- To analyze IT support ticket data to understand performance trends
- To identify bottlenecks and inefficiencies in ticket resolution
- To provide data-driven insights to improve IT support team performance
- To recommend strategies for optimizing workflow and resource allocation
- To enhance overall service quality and customer satisfaction

Tools Used:

- Jupyter NoteBook(Pandas, NumPy),PowerBI

Data Preprocessing (Milestone-1):

Data cleaning, handling missing values, and feature creation were performed.

Exploratory Data Analysis (Milestone-2):

Visualizations were created to analyze ticket distribution, priority, categories, and regions.

Performance Analysis (Milestone-3):

Performance trends were analyzed based on resolution time, priority, product, and region.

Key Insights:

- The majority of tickets are related to technical and transactional issues
- A significant number of tickets remain open or pending
- Certain clusters show repeated issues (high similarity)
- Some products generate higher ticket volumes
- Most tickets are resolved quickly, but slow-resolution cases need improvement
- No major difference in performance across countries
- High-priority tickets are not resolved significantly faster than others
- Workload is evenly distributed across region

Conclusion:

The project successfully analyzed IT support performance and provided actionable insights to improve efficiency and service quality.